G MUKUND

mukund.ganesh2002@gmail.com +91 7217761576 GMukund2002 mukundganesh

Education

BTech in Computer Science and Engineering, *Indian Institute of Technology (IIT), Jodhpur*, GPA: 7.58/10 2021 — 2025

Work Experience

Centific **May 2025 — Present**
Junior AI Engineer *Hyderabad, India*

- Built and productionized a real-time, multi-stream **Agentic Vision AI platform** using **DeepStream**, **TensorRT**, **Docker** and **Kubernetes**, combining event-driven analytics with **LLM, VLM and rule-based agent workflows** for autonomous incident understanding, and action orchestration across use cases (drone human behavior, vehicle LPR, and perimeter breach).
- Developed **VisionAI microservices** with **REST APIs** for pipeline orchestration (lifecycle, health, ROI), streaming analytics via **Kafka**, and publishing unified alerts via **RabbitMQ** (alert fan-out, queueing) with resilient, asynchronous inference workflows.
- Implemented scalable inference pipelines combining **YOLO**, **RT-DETR**, and agentic reasoning, with **schema validation, retries, fault tolerance, and observability** using **Grafana, ELK, and Kibana**.
- Built semantic intelligence layers using **CLIP** + **NVIDIA Cosmos VLM** + **PaddleOCR** for cross-camera de-duplication, incident similarity search, high-accuracy OCR, and automated agent-generated incident summaries with contextual reasoning.

Internship Experience

Technical University of Munich **May 2024 — August 2024**
Research Intern *Munich, Germany*

- Modeled beta coalescent tree sequences with varying alpha values using **Graph Neural Networks (GNNs)** and **Graph Attention Networks (GATs)** to capture structured relationships and demographic transition dynamics through transition matrices (TSMC).
- Built a **Vision Transformer (ViT) pipeline** over **10×10 transition-matrix patches** with positional embeddings and Transformer encoders for and population size prediction, achieving stronger performance for higher alpha values.

International Institute of Information Technology Bangalore **June 2024 — October 2024**
Research Intern -> Research Assistant *Bangalore, India*

- Developed a 3D visual grounding pipeline on **ReferIt-3D** dataset by extracting subject-object relationships from utterances using a custom NLP approach and generating embeddings via **BERT** to identify and represent competing sentences.
- Curated a competing-scenes dataset with **OpenCV**-based masking of 3D point-clouds to improve training on ambiguous cases.
- Architected a **VLM** combining **PointNet++**, **BERT** embeddings, with custom **contrastive loss** for resolving competing 3D scenes.

Indian Institute of Technology Indore **June 2023 — August 2023**
Research Intern *Indore, India*

- Engineered low-resource **neural machine translation(NMT)** systems for Indian and Sorbian language pairs using Transformers, BERT, and seq2seq models. Used **IndicTrans** and **AI4Bharat** with web-scraped, OCR-derived corpora, data augmentation, and hyperparameter tuning to improve translation fidelity.

Research Projects

NLP for Bridging Ontologies of Ayurveda and Modern Sciences **August 2024 — January 2025**

- Fine-tuned **BERT** on Ayurvedic and modern medical concepts for **ontology-aware embeddings**, performed two-way description alignment using **cosine similarity**, and integrated the retrieval layer into a **RAG pipeline** for knowledge grounding.
- Fine-tuned **LLaMA 3** on Charaka Samhita and PubMed articles to build a domain-specific generative QA system combining Ayurvedic and modern medical knowledge for context-driven response generation.

Autonomous Multiagent Campus Virtual Tour | GitHub **September 2024 — October 2024**

- Engineered a **multi-agent system** for autonomous campus tours using **ROS, CrewAI, and RViz**, enabling real-time communication and visualization between campus management and visitors through custom **message protocols**.
- Designed dynamic agent-role selection, path optimization, and error-handling to improve navigation reliability and interaction.

Technical Skills

Languages: Java, Python, C++, C, R, JavaScript, HTML, CSS, Bash, LaTeX
AI/ML: PyTorch, TensorFlow, Hugging Face, OpenCV, MySQL, PostgreSQL, Kafka, RabbitMQ, FastAPI, REST APIs
Agentic tools: dsPy, LangGraph, LangChain, Claude Code, OpenCode
DevOps & SWE tools: Docker, Kubernetes, Grafana, Kibana, ELK, Prometheus, GraphQL, ReactJS, NodeJS , Git, PyTest

Conferences, Publications and Achievements

AdvPred-Core-vs-FocusPred-Non-Attention-vs-Attention ICCIML, Springer 2024
Teaching Assistant at IIT Jodhpur - Natural Language Processing Spring 2025